THE MALAŴI NATIONAL EXAMINATIONS BOARD

2006 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/I

Time Allowed: 2 hours

Tuesday, 17 October **8:30 – 10:30 am**

PAPER I

(100 marks)

Instructions

1. **This paper contains 6 pages. Please check.**
2. Answer **all** the **24** questions in this paper.
3. The maximum number of marks for each answer is indicated against each question.
4. Mathematical tables, graph paper and answer books are provided.
5. Calculators may be used.
6. Used graph paper and/or supplementary sheets must be handed in together with the answer book.
7. **All working must be clearly shown**; it should be done on the same sheet as the rest of the answers.
8. Write your **Examination Number** on top of each page of your Answer Book.

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Answer **all** the **twenty-four** questions.

1. Simplify the following fraction:

$$\frac{3x+6}{(x-1)(x+2)}.$$ **(3 marks)**

2. Factorise completely $2x^2 + 4xy - 30y^2$. **(3 marks)**

3. A cuboid is 76 cm long, 50 cm wide and 40 cm high. Calculate the volume of the cuboid. **(3 marks)**

4. If $f(x) = 8^x - 6$, find $f\left(\frac{2}{3}\right)$. **(4 marks)**

5. **Figure 1** is a triangle **ABC** in which angle **BAC** = 60°, **AC** = 10 cm and the area of the triangle is $15\sqrt{6}$ cm^2.

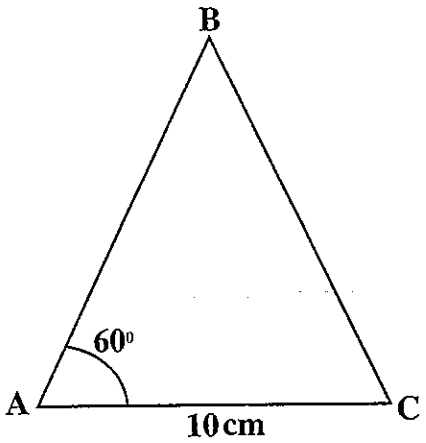

Figure 1

Calculate the length of **AB** leaving the answer in its simplest surd form. **(5 marks)**

6. When the polynomial $ax^2 + bx + c$ is divided by $(x + 1)$ and $(x + 3)$ gives a remainder of 2 in each case. Find the polynomial. **(4 marks)**

Continued/...

7. **Table 1** shows marks that student **A** and student **B** got from tests. Student **A** sat for 5 tests while student **B** sat for 4 tests. Student **B** has mark x missing.

Table 1

STUDENT A	55	70	80	30	65
STUDENT B	67	60	x	53	

Given that the mean mark of student **A** is the same as the mean mark of student **B**, calculate the value of x. **(5 marks)**

8. John is twice as old as Mary. If the sum of the squares of their ages is 125, how old is Mary? **(6 marks)**

9. Without using a calculator or four-figure tables, find Cos B if Sin B = 0.8. **(6 marks)**

10. Triangle **ABC** has vertices **A**(-1,2), **B**(3,3) and **C**(2,-1). Prove that angle **BAC** = angle **ACB**. **(5 marks)**

11. Find the values of x and y in the following matrix equation:

$$\begin{pmatrix} 6 & 3 \\ 2 & -3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12 \\ -4 \end{pmatrix}.$$ **(5 marks)**

12. **Table 2** shows the speed of a train recorded every 10 seconds.

Table 2

SPEED (m/s)	200	400	600	800
TIME (s)	10	20	30	40

Using a scale of 2 cm to represent 100 m/s on the vertical axis and 2 cm to represent 10 seconds on the horizontal axis, draw a speed-time graph and use it to calculate the acceleration of the train. **(6 marks)**

13. Given that $x \propto \frac{y}{z}$. When $x = 10$, $y = 2$ and $z = 4$. Find value of x when $y = 1$ and $z = 5$. **(5 marks)**

Continued/...

14. Two lines **G** and **H** intersect at a point **P**. **G** passes through the points (–4, 0) and (0,6). Given that **H** has the equation: $y = 4x - 4$, find, by calculation, the coordinates of **P**. **(3 mar**

15. **Figure 2** shows a venn diagram representing set of all numbers (**N**), set of even numbers (**E**), set of odd numbers (**0**) and set of prime numbers (**P**). s)

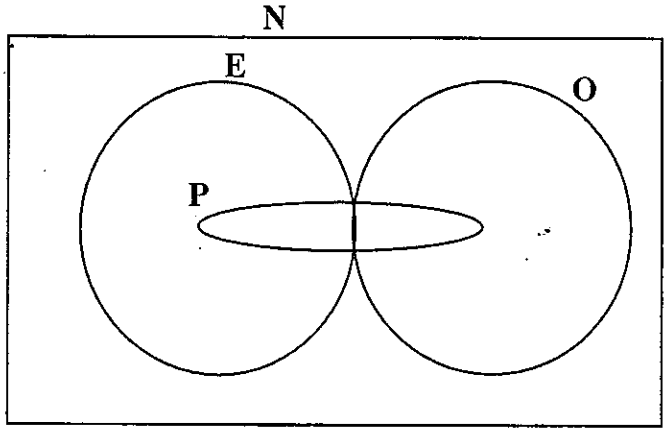

Figure 2

Copy the venn diagram and place the numbers $\frac{1}{2}$, 2, 6, 9 and 13 in the right places. **(6 marks)**

16. Using a ruler and pair of compasses only, draw a line **AB** = 10 cm and construct a circle with the line **AB** as a diameter. Mark a point **C** on the circle such that **AC** = 6 cm. Join **AC** and **BC**. Construct a tangent **CP** such that angle **BCP** is acute. **(5 marks)**

17. In **Figure 3**, circle **ABCD** has chords **AC** and **BD** intersecting at **H**. Length of line **HD** is twice that of **BH**.

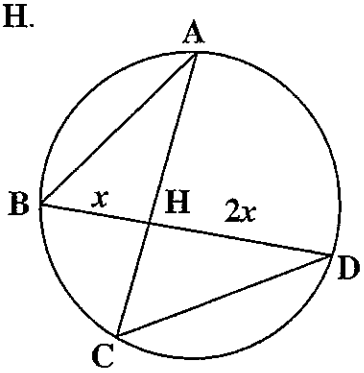

Figure 3

$a(-3)^2 + b(-3) + c = 2$

$9a - 3b + c = 2$

$9a - 3b + c = 2$

Find the ratio of areas of triangle **ABH** to triangle **CDH.** **(4 marks)**

$a(-1)^2 + b(-1) + c = 2$

$a - b + c = 2$

$-(9a - 3b + c = 2)$

$a - b + c = 2$

$8a - 2b = 0$

Continued/...

igure 4 shows graphs of $x + y = 8$ and $x - 2y + 4 = 0$.

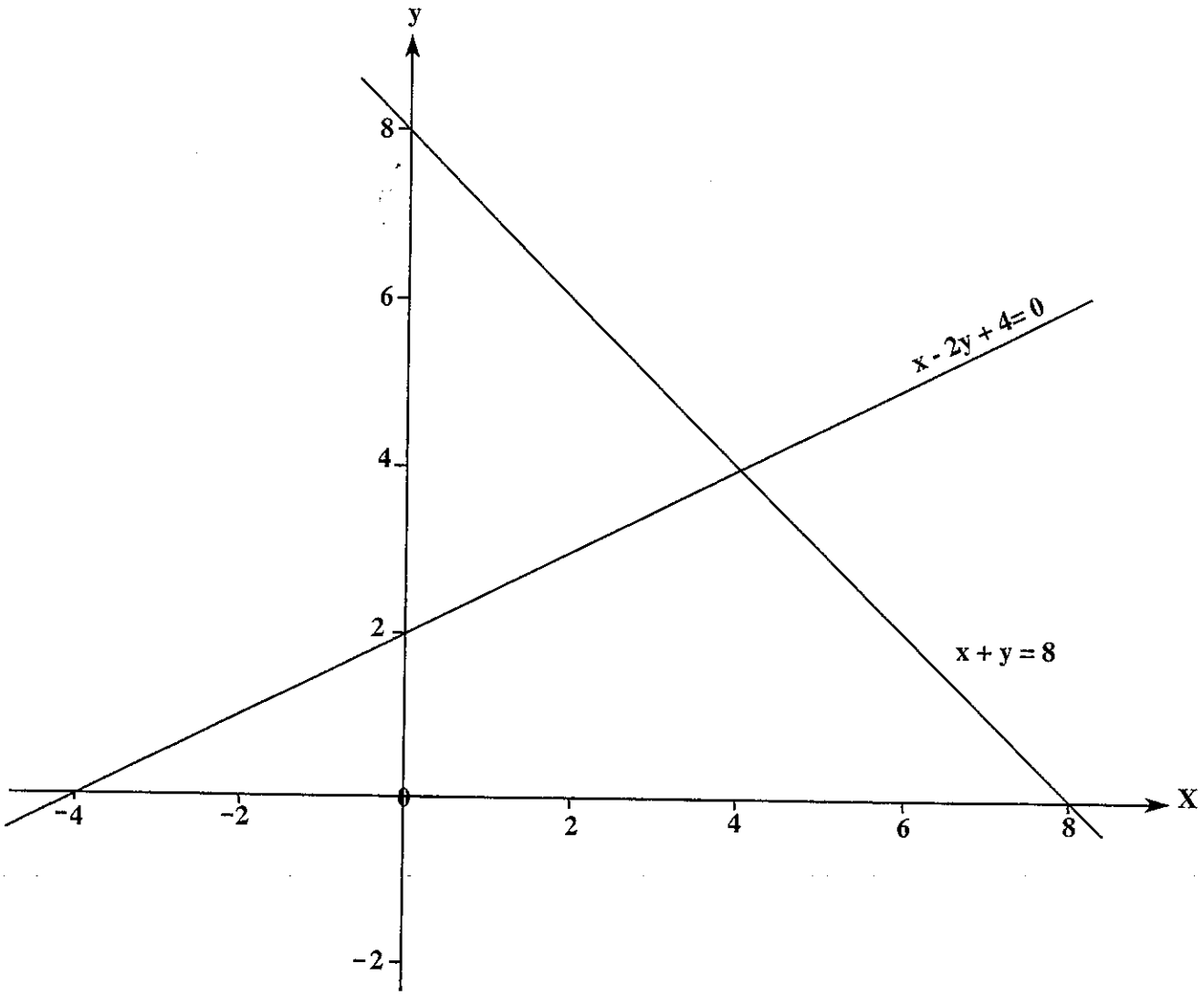

Figure 4

Copy the figure on the graph paper provided and show the region bounded by the following inequalities;

$$x \geq 0,$$
$$x + y \leq 8,$$
$$x - 2y + 4 \geq 0,$$

by shading the unwanted region. **(6 marks)**

Continued/...

19. **Figure 5** shows a circle **ABCD** in which **AB** and **DC** are produced to **G** and **F** respectively such that **GF** is parallel to **AD**.

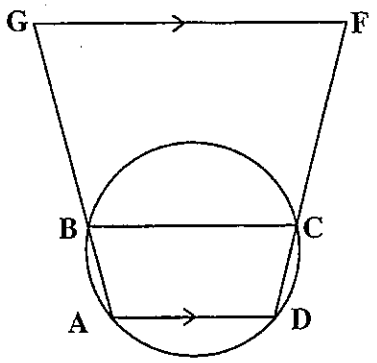

Figure 5

Prove that quadrilateral **GBCF** is cyclic. **(4 marks)**

20. Given that $\log_{10} n - \log_{10} m = 2 \log_{10} h$, show that $n = mh^2$. **(4 marks)**

21. Coin **A** is tossed followed by coin **B**. The probability that coin **A** shows head is $\frac{1}{2}$ while the probability that coin **B** shows head is $\frac{1}{4}$. Using a tree diagram, calculate the probability that both coins **A** and **B** show tails. **(4 marks)**

22. Find the sum of the first 12 terms of the following **GP**:

$$\frac{1}{2187}, \frac{1}{729}, \frac{1}{243}, \ldots$$

Give your answer correct to 2 decimal places. **(5 marks)**

23. Simplify $\dfrac{x^2-2}{x-\sqrt{2}}$. **(3 marks)**

24. **Figure 6** shows a parallelogram **ABCD** in which $\overrightarrow{AB} = \underline{x}$ and $\overrightarrow{BC} = \underline{y}$.

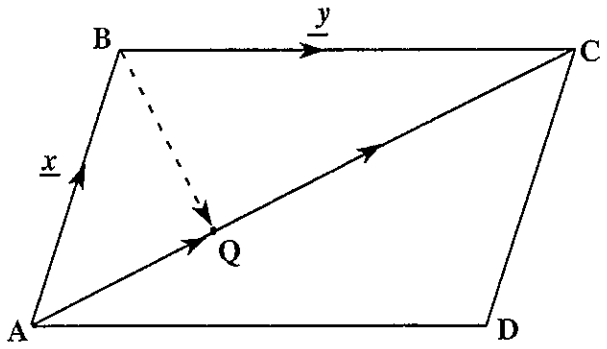

Figure 6

If $\overrightarrow{AQ} = \frac{1}{4}\overrightarrow{AC}$, find $\overrightarrow{BQ}$ in terms of $\underline{x}$ and $\underline{y}$. **(5 marks)**

END OF QUESTION PAPER

NB: This paper contains 6 pages.